

NEW GROWN DIAMOND

NGD Project Work Report

Technical, design and production-readiness review of the animated B2B lab-grown diamond platform.

Executive summary

NGD has developed from a basic Vite starter into a polished, responsive diamond inventory and lead-generation platform. Its strongest qualities are premium art direction, purposeful animation, accessibility work and real business workflows. The main work remaining is technical SEO, route-level performance optimization, blog article routing, documentation and stronger US wholesale positioning.

AREA	ASSESSMENT	STATUS
Visual design	Premium, distinctive luxury presentation	Excellent
Animation	Scroll, parallax, transitions, canvas and reduced-motion support	Very good
Responsive design	Validated across five viewport sizes	Very good
B2B functionality	Inventory, enquiries, offices, certificates and admin workflow	Strong foundation
Accessibility	Semantic structure, keyboard support and ARIA coverage	Good
SEO	Core metadata, structured data and prerendering are incomplete	Needs work
Production readiness	Main features built; optimization and deployment validation remain	Approximately 70%

Technology and architecture

- React 19
- Vite 8
- CSS Modules
- GSAP
- Framer Motion
- Lenis
- Three.js
- Supabase
- Vercel / Netlify

The application uses a custom client-side router and reusable page, motion, media, chrome and product components. The codebase currently contains 25 page files, 55 component files, 25 section files, 6 hooks and 17 library files.

Completed work

Premium frontend Animated homepage, smooth scrolling, pointer parallax, page transitions, custom diamond cursor, preloader, dynamic backgrounds, video storytelling and dark/light themes.	Public website Home, diamonds, jewellery, about, education, shapes, why lab-grown, FAQ, contact, blogs and custom not-found experiences.
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Inventory

Live Supabase stock, dynamic shape filters, diamond cards, specifications, photography, certificates, video/360 viewing and linked product enquiries.

Accounts and admin

Registration, login, password recovery, profiles, protected admin routes, diamond administration and configuration-aware states.

Lead generation

Structured B2B enquiry form with company, country, product reference, enquiry type and detailed diamond requirements, plus spam protection.

International positioning

Manufacturing and certification content plus office information for Surat, Mumbai, New York and Hong Kong.

Accessibility and responsive work

- Semantic heading hierarchy and descriptive image alternatives.
- ARIA labels, live regions, loading states and button state attributes.
- Keyboard-accessible mobile navigation with focus trapping, Escape handling and focus restoration.
- Reduced-motion alternatives across the animation system.
- Responsive screenshots for 320×568, 375×812, 768×1024, 1440×900 and 1920×1080.

Quality verification

CHECK	RESULT	NOTES
Production build	Passed	194 modules transformed successfully.
Lint	Passed with one warning	State update inside an effect in ContinueNext.jsx.
Main JavaScript	121 KB gzip	Route splitting can reduce initial delivery.
Supabase bundle	54 KB gzip	Currently included in the production output.
GSAP bundle	30 KB gzip	Reasonable for the animation-heavy experience.
CSS	31 KB gzip	Single large stylesheet output.
Largest video	1.33 MB	Requires careful preload and mobile delivery strategy.

Outstanding risks

1. Blog article routes

The journal creates links in the form `/blogs/post-slug`, but the application currently handles only the exact `/blogs` route. A blog detail page and dynamic slug route are required.

2. Technical SEO

Route-specific descriptions, canonical URLs, Open Graph metadata, Twitter cards, robots.txt, sitemap.xml, structured data and prerendered public pages are not yet implemented. Admin and account routes also need noindex rules.

3. Performance

Public, authentication and admin pages are imported eagerly. React lazy loading and route-based code splitting should keep Supabase, admin, blog and inventory code out of the initial homepage download.

4. US B2B positioning

A dedicated US wholesale landing page should explain USD quotations, minimum orders, insured shipping, lead times, inspection/return terms, matching pairs, calibrated parcels, private-label services and New York appointments.

5. Documentation and source control

The README is still the default Vite document. Most implementation files are currently uncommitted or untracked, while the repository contains only the initial project commit. Documentation and a reviewed checkpoint commit are needed.

Recommended priority plan

1. Implement individual blog detail pages and slug routing.
2. Add route-specific SEO metadata, canonical URLs and social previews.
3. Create robots.txt, sitemap.xml and schema.org structured data.
4. Prerender public marketing routes and provide correct HTTP 404 behavior.
5. Add route-level lazy loading and optimize heavy media delivery.
6. Create a dedicated USA wholesale landing page and stronger US CTAs.
7. Test live inventory, authentication and enquiry workflows end to end.
8. Replace the starter README with complete setup and deployment documentation.
9. Review environment files and create a clean source-control checkpoint.

Conclusion

NGD is no longer simply an animated landing page. It is a substantial luxury B2B web platform with a real inventory, customer account, enquiry and administration foundation. The design work is the project's clearest advantage. Completing SEO, performance optimization, content routing and operational documentation will move it from a strong showcase toward a production-ready international sales platform.

This report is based on a read-only review of the ngd workspace, production build output, lint results and stored responsive screenshots. No application source files were changed during the audit.